

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a53e71c67143a43aa.jsonl`  
- SHA-256: `a4241f01d3a529d1865d61991ec34be9887e427172b6521edc08b1e82a24c487`  
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- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:23:54.321Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022-2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 - ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96-99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **recent** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"ASRF (Action Segment Refinement Framework) reduces over-segmentation in temporal action segmentation via a dual-branch design: an Action Segmentation Branch (ASB) that classifies frames, plus a Boundary Regression Branch (BRB) that regresses action boundary probabilities, whose detected boundaries refine the ASB output."

****Source:**** <https://arxiv.org/abs/2007.06866> (primary)

****Supporting quote:**** "a long-term feature extractor and two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB) ... The action boundaries predicted by the BRB refine the output from the ASB, which results in a significant performance improvement."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:23:56.350Z)

I'll verify this claim about ASRF's architecture by examining the primary source.

3. assistant (2026-06-16T03:23:59.096Z)

I'll fetch the primary source to verify the architectural claim about ASRF.

4. user (2026-06-16T03:24:06.713Z)

ASRF Framework Architecture

Yes to all your questions. Here's the framework breakdown:

(1) Framework Name: The paper proposes "an Action Segment Refinement Framework (ASRF)."

(2) Two-Branch Design: The model includes "the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB). The ASB classifies video frames with action classes, while the BRB regresses the action boundary probabilities."

(3) Refinement Process: "The action boundaries predicted by the BRB refine the output from the ASB, which results in a significant performance improvement."

(4) Feature Extractor: The architecture contains "a long-term feature extractor" that "provides shared features for the two branches with a wide temporal receptive field."

(5) Over-segmentation Goal: The paper directly addresses this in its title: "Alleviating Over-segmentation Errors by Detecting Action Boundaries." The framework achieves substantial gains, with improvements of up to 13.7% in segmental edit distance and up to 16.1% in segmental F1 score across three datasets.

5. assistant (2026-06-16T03:24:08.798Z)

The primary source confirms the claim in full detail. Let me do one quick cross-check to ensure source quality and verify the publication details.

6. user (2026-06-16T03:24:17.409Z)

Web search results for query: "ASRF Action Segment Refinement Framework Ishikawa boundary regression branch over-segmentation WACV 2021"

Links: [{"title": "WACV 2021 Open Access Repository", "url": "https://openaccess.thecvf.com/content/WACV2021/html/Ishikawa_Alleviating_Over-Segmentation_Errors_by_Detecting_Action_Boundaries_WACV_2021_paper.html"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://yiskw713.github.io/asrf/"}, {"title": "[PDF] Alleviating Over-segmentation Errors by Detecting Action Boundaries | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Alleviating-Over-segmentation-Errors-by-Detecting-Ishikawa-Kasai/4fed0235614df46fa0c53e05b71e9939b53a1bdf"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://arxiv.org/pdf/2007.06866"}, {"title": "GitHub - yiskw713/asrf: Code for 'Alleviating Over-segmentation Errors by Detecting Action Boundaries' accepted in WACV2021", "url": "https://github.com/yiskw713/asrf"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | Request PDF", "url": "https://www.researchgate.net/publication/352386005_Alleviating_Over-segmentation_Errors_by_Detecting_Action_Boundaries"}, {"title": "Alleviating over-segmentation errors by detecting action boundaries ? Keio University", "url": "https://keio.pure.elsevier.com/en/publications/alleviating-over-segmentation-errors-by-detecting-action-boundari-2"}, {"title": "Alleviating Over-Segmentation Errors by Detecting Action Boundaries", "url": "https://papertalk.org/papertalks/27315"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://openaccess.thecvf.com/content/WACV2021/papers/Ishikawa_Alleviating_Over-Segmentation_Errors_by_Detecting_Action_Boundaries_WACV_2021_paper.pdf"}, {"title": "Uncertainty-Aware Representation Learning for Action"}]

Segmentation", "url": "https://www.ijcai.org/proceedings/2022/0115.pdf"}]

Based on the search results, I found comprehensive information about the ASRF (Action Segment Refinement Framework) paper:

ASRF: Action Segment Refinement Framework (WACV 2021)

The Action Segment Refinement Framework (ASRF) is an effective framework for the temporal action segmentation task, authored by Yuchi Ishikawa, Seito Kasai, Yoshimitsu Aoki, and Hirokatsu Kataoka.

Architecture

The model architecture consists of a long-term feature extractor and two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB). The ASB classifies video frames with action classes, while the BRB regresses the action boundary probabilities.

Key Contribution

The action boundaries predicted by the BRB refine the output from the ASB, which results in a significant performance improvement. The ASB and the BRB are complementary to each other, which enables the reduction of over-segmentation errors.

Performance

The model outperforms state-of-the-art methods on three challenging datasets, offering an improvement of up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score.

The official implementation code is publicly available on GitHub (yiskw713/asrf), and the paper was published in the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2021.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. assistant (2026-06-16T03:24:26.739Z)

The claim is fully corroborated by both the primary arXiv source and multiple independent sources (CVF WACV 2021 official proceedings, official GitHub repo, Semantic Scholar, Keio University publication record).

Verification against the checklist:

1. ****Supported by quote?**** Yes, exactly. Every component of the claim maps verbatim: ASRF = Action Segment Refinement Framework; dual-branch with ASB (frame classification) + BRB (boundary probability regression); BRB boundaries refine ASB output; long-term feature extractor providing shared features. The supporting quote is accurate.
2. ****Contradicting evidence?**** None found. Multiple independent sources (CVF official proceedings, GitHub yiskw713/asrf, Semantic Scholar, Keio University) all describe the identical architecture and contribution.
3. ****Source quality vs claim strength?**** Primary peer-reviewed source (WACV 2021, IEEE/CVF) plus the arXiv preprint and official code. This is a strong, factual architectural description, not an extraordinary claim ? source quality exceeds what is required.
4. ****Outdated?**** Published 2020 (arXiv) / WACV 2021. The claim is purely descriptive of a static published architecture ? descriptions of an architecture do not "go stale." Still actively cited and used as a baseline.
5. ****Marketing/cherry-picked?**** No. Peer-reviewed academic paper; the architectural description is verifiable and standard.

One minor note on precision: the BRB regresses *boundary probabilities* (a probability per frame of being a boundary), which the claim phrases as "regresses action boundary probabilities, whose detected boundaries refine the ASB output." This is an accurate characterization ? the BRB produces a boundary probability sequence, boundaries are then detected from it, and those refine ASB. No overreach.

8. user (2026-06-16T03:24:32.552Z)

Structured output provided successfully

9. assistant (2026-06-16T03:24:34.695Z)

Verification complete. The claim is ****not refuted**** ? it is fully supported by the primary peer-reviewed source (arXiv:2007.06866 / WACV 2021, Ishikawa et al.) and corroborated by multiple independent sources, with high confidence.